A CONSTRUCTIVE SERRE CRITERION FOR AFFINENESS
A DETAILED INFORMAL PROOF FOR LOCALLY RINGED LATTICES

Abstract. We give a point-free proof that a qcqs scheme whose quasi-coherent modules have vanishing first cohomology is affine. For a finite affine cover, add its support ideals and consider the fibre of their addition map over 1. Its translation module is a saturated submodule of a finite product of support ideals, hence quasi-coherent. One application of the torsor hypothesis gives a subordinate partition of unity. The resulting global principal affine cover and the same unit equation give the final affineness comparison. Canonical localisation maps are defined before models; their equivalence descends through finite sheaf equalisers. For the converse, the canonical maps of a quasi-coherent module are equivalences on every open, by descent from the charts of an atlas; this reconstructs the module on every affine open and reduces torsor triviality to degree-one gluing. Vanishing of Čech H^1 on one finite affine cover is equivalent to its vanishing on every such cover and to affineness, and the groups of two such covers are canonically isomorphic, through torsors. All of this is checked in the Unit development `unit/sag/qcqs` (Section 10).

1. Scope, conventions, and the theorem

The geometric framework is the existing qcqs development in Unit. A scheme is a small locally ringed distributive lattice $X = (L, \mathcal{O})$ which merely has a finite affine open cover. The lattice is bounded; its elements represent compact opens, its bottom is 0, and its top is 1. All rings are unital commutative rings with set carriers; the zero ring and the empty scheme are included. Here and below "scheme" means qcqs scheme in this sense.

For $a \in L$ and $t \in \mathcal{O}(a)$, write $D_a(t)$ for the greatest $b \leq a$ on which t is invertible. Write $D(f) = D_1(f)$ for a global section. Restriction is denoted $t|b$ when the source is clear. Comparisons between different presentations of a ring or module use their specified isomorphisms. An expression such as $\mathcal{O}(D_A(h)) = A_h$ below abbreviates its canonical comparison, not a judgemental equality. All existential choices of presentations, denominators, covers, and lifts are mere: the displayed constructions are first made for actual witnesses and then eliminated only into propositions or uniquely determined data. Finite choice for propositional truncation suffices.

Theorem 1.1 (External constructive Serre criterion)**.** *Let X be a small qcqs locally ringed lattice. Suppose that for every small quasi-coherent $\mathcal{O}_X$-module M, every M-torsor merely has a global section. Then the canonical map*

$$X \longrightarrow \operatorname{Spec} \mathcal{O}_X(1)$$

is an isomorphism. Conversely, if X is affine, every torsor under a small quasi-coherent module merely has a global section. Equivalently, X is affine if and only if $\check{\mathrm{H}}^1(\mathcal{U}, M) = 0$ for every small quasi-coherent M, where $\mathcal{U}$ is one, or equivalently every, finite affine cover (Section 5). For any X with a quasi-coherent M, the groups of two finite affine covers are canonically isomorphic.

The direct implication uses the atlas-free torsor hypothesis as stated. Its route is support ideals, a partition torsor, one application of the hypothesis, and localisation descent for the

Date: October 30, 2026.

structure sheaf. Associated exactness, sheaf cokernels, general kernel closure, affine reconstruction and the results of Section 5 are not prerequisites. Section 4 provides the canonical localisation maps and a single criterion for associated affine models. The converse (Corollary 5.3) and the H^1 form (Corollary 5.5) are separate consequences.

The proof uses no higher cohomology, closed points, prime ideals, generalized field axiom, synthetic scheme axiom, or resizing principle. The original affine-replacement argument supplied in the conversation was credited there to a discussion involving David and Felix. The partition-of-unity proof below replaces its successive lifting steps by one simultaneous lift. This note makes no further claim about priority or the identities of the other participants.

2. Geometric facts and finite affine refinements

The following are interfaces already established in the geometric development. They must be used by their names in the formal proof.

$$D_b(t|b) = b \wedge D_a(t) \qquad (b \leq a), \tag{1}$$

$$D_a(1) = a, \qquad D_a(0) = 0, \tag{2}$$

$$D_a(st) = D_a(s) \wedge D_a(t), \qquad D_a(s+t) \leq D_a(s) \vee D_a(t). \tag{3}$$

Also $D_a(-t) = D_a(t)$, since -1 is a unit. A section t is invertible in $\mathcal{O}(a)$ if and only if $D_a(t) = a$. For a finite cover $a = \bigvee_i a_i$, the sheaf condition for units gives

$$D_a(t) = \bigvee_i D_{a_i}(t|a_i). \tag{4}$$

These formulas concern invertibility support; they do not assert that it is the universal support of $\mathcal{O}(a)$ for a general a.

On an affine open w, with $B = \mathcal{O}(w)$, the canonical comparison identifies the downset of w with $\mathrm{Zar}(B)$. Every $b \leq w$ is therefore merely a finite join of $D_w(h_i)$. Every such principal open is affine, and its ring of sections is canonically B_{h_i}. In particular every compact open of a qcqs scheme merely has a finite affine cover: intersect it with a finite affine atlas and refine each intersection inside that atlas member.

Lemma 2.1 (Common refinement)**.** *Two finite affine covers of a compact open merely admit a common finite affine refinement, with specified inclusions into members of each cover.*

Proof. Intersect every pair of members. Each intersection is compact and is an open of the first affine member, so express it as a finite join of principal opens in that member. These are affine, lie in both original members, and jointly cover the intersection. Collect the finitely many finite families. This does not assert that the intersections themselves are affine. □

One can also begin with the formulation in the question: a ringed lattice with a finite affine cover, without including invertibility support in its data. For a section $t \in \mathcal{O}(a)$, intersect that cover with a and refine inside each affine chart by finitely many principal opens. On each resulting affine piece the restricted section has an invertibility open. Their finite join is an invertibility open on a: a restriction is a unit exactly when its restrictions on this finite cover are units, since inverses glue uniquely. Uniqueness of the greatest such open makes the construction independent of all refinements and presentations. The support laws are checked on the affine pieces and then by the covering equation. Thus invertibility support is uniquely recoverable from this affine-local condition. For the subsequent proof we use the existing locally ringed record.

3. Elementary module localisation

The direct route uses fractions, finite sums and the finite-equaliser localisation descent step. Cokernel exactness and degree-one gluing belong to the broader algebraic and converse results recorded here.

For an A-module N, write N_f for localisation at the powers of f. Its elements are represented by n/f^r. Equality of two fractions means merely that an additional power of f annihilates their cross-multiplied difference. In particular

$$n/1 = 0 \text{ in } N_f \quad \Longleftrightarrow \quad \exists r \geq 0 \; f^r n = 0. \tag{5}$$

All maps of localised modules below are specified by their value on $n/1$ and the inversion of f; their composition equations follow by uniqueness, not by re-expanding fractions.

Lemma 3.1 (Finite exactness)**.** *Localisation of modules preserves finite products, kernels, and cokernels, and hence finite equalisers and short exact sequences.*

Proof. A finite tuple of fractions can be put over a common denominator, giving the inverse to the product comparison. For a map $v : N \to P$, if $v(n)/f^r = 0$, equation (5) gives $f^e v(n) = 0$. The fraction n/f^r can then be written $(f^e n)/f^{r+e}$ with numerator in $\ker v$. Injectivity follows from the same equality criterion. For cokernels, lift a representative of a quotient fraction to N and use the fraction equality criterion to identify its kernel with the image of the localised map. All these representative choices are mere, and the induced maps are unique. □

Lemma 3.2 (Powers of a unit-ideal family)**.** *If $(f_1, \ldots, f_m) = A$, then $(f_1^r, \ldots, f_m^r) = A$ for every positive r.*

Proof. The Zariski support has $D(f_i^r) = D(f_i)$, so the join of these opens is top. The formal Nullstellensatz says that 1 belongs to the ideal of these powers. Equivalently, expand a sufficiently large power of an equation $1 = \sum_i a_i f_i$ and group terms by an index appearing at least r times. Thus coefficients λ_i with $\sum_i \lambda_i f_i^r = 1$ merely exist. □

Lemma 3.3 (Degree zero gluing)**.** *If $(f_1, \ldots, f_m) = A$, then*

$$N \longrightarrow \prod_i N_{f_i} \rightrightarrows \prod_{i,j} N_{f_i f_j}$$

is an equaliser.

Proof. For injectivity, a difference which vanishes in every N_{f_i} is killed by a power of each f_i. Choose one positive bound r for these finitely many powers. Lemma 3.2 then annihilates the difference by 1.

For existence, let $s_i \in N_{f_i}$ be compatible. Choose a common positive r and $t_i \in N$ with $t_i|f_i = f_i^r s_i$. For every i, j, the difference $t_i|f_j - f_i^r s_j$ vanishes after further localising at f_i. Choose a common killing exponent e for these finitely many differences and replace t_i by $f_i^e t_i$ and r by $r + e$. Now $t_i|f_j = f_i^r s_j$ in N_{f_j} for all i, j. Choose λ_i with $\sum_i \lambda_i f_i^r = 1$ and set $t = \sum_i \lambda_i t_i$. Its image in N_{f_j} is s_j. Uniqueness has already been proved, so the mere representative choices produce the uniquely determined amalgam. □

Lemma 3.4 (Gluing over a basic open)**.** *Suppose $D_A(h_i) \leq D_A(g)$ and $D_A(g) \leq \bigvee_i D_A(h_i)$. Then*

$$N_g \longrightarrow \prod_i N_{h_i} \rightrightarrows \prod_{i,j} N_{h_i h_j}$$

is an equaliser, with the canonical localisation restrictions.

Proof. The denominator and error clearing in Lemma 3.3 take place in the original module and its localisations, before the partition of one is used. They therefore give one exponent r and numerators $t_i \in N$ satisfying $t_i|h_j = h_i^r s_j$ for compatible local data $s_j \in N_{h_j}$. Since $D_A(h_i) \leq D_A(h_i^r)$, the formal Nullstellensatz gives, merely,

$$g^e = \sum_i \lambda_i h_i^r.$$

Put $t = \sum_i \lambda_i t_i$ and take $s = t/g^e \in N_g$. In N_{h_j} we have $t|h_j = g^e s_j$, and g acts invertibly there because $D_A(h_j) \leq D_A(g)$. Hence $s|h_j = s_j$.

For uniqueness, represent a difference by x/g^q. Vanishing on every N_{h_i} implies that powers of the h_i kill x. The formal Nullstellensatz expresses a power of g as a linear combination of those killing powers. Thus a power of g kills x, so $x/g^q = 0$. All choices are eliminated into equality or the proposition that the amalgam exists uniquely. The argument includes an empty covering family and the zero ring. □

Degree-one gluing for the converse. This final algebraic lemma is used for affine torsor triviality, not for the direct implication of the criterion.

Lemma 3.5 (Degree one gluing)**.** *If $\bigvee_i D_A(f_i) = 1$, every family $c_{ij} \in N_{f_i f_j}$ satisfying*

$$c_{ij}|f_k + c_{jk}|f_i = c_{ik}|f_j \quad \text{in } N_{f_i f_j f_k}$$

merely admits $b_i \in N_{f_i}$ with $b_j - b_i = c_{ij}$ on every overlap.

Proof. Choose a common positive r and elements $a_{ki} \in N_{f_i}$ whose images in $N_{f_k f_i}$ are $f_k^r c_{ki}$. This is possible by clearing the f_k denominators in the finitely many c_{ki}. In $N_{f_i f_j}$ put

$$e_{kij} = a_{kj}|f_i - a_{ki}|f_j - f_k^r c_{ij}.$$

After localising at f_k this is zero by the cocycle equation. Choose one exponent e killing every e_{kij} by the appropriate f_k^e. Replace a_{ki} by $f_k^e a_{ki}$ and r by $r + e$. We now have the actual equations

$$a_{kj}|f_i - a_{ki}|f_j = f_k^r c_{ij} \quad \text{in } N_{f_i f_j}.$$

The lattice covering equation, the equality $D_A(f_k^r) = D_A(f_k)$ and the formal Nullstellensatz give, merely, λ_k with $\sum_k \lambda_k f_k^r = 1$. Set $b_i = \sum_k \lambda_k a_{ki}$. Subtracting the two expressions on an overlap gives $b_j - b_i = c_{ij}$. All choices remain inside the mere assertion that a coboundary exists. □

For an empty covering family, A is the zero ring and every unital A-module is zero. Both lemmas then assert contractibility of the appropriate empty diagram, and hold as well.

4. Module sheaves and quasi-coherence

Definition 4.1. A module sheaf M on X assigns an $\mathcal{O}(a)$-module $M(a)$ to each $a \in L$, with additive restriction maps which respect scalar restriction. It satisfies the finite-cover sheaf condition on the lattice. A morphism is a natural family of linear maps. It is locally surjective if for every $y \in F(a)$ there merely exist a finite cover $a = \bigvee_i a_i$ and $x_i \in E(a_i)$ mapping to $y|a_i$.

For an A-module N, define $\widetilde{N}$ on $\operatorname{Spec} A$ directly: on an open w, a section is a compatible family of elements of N_h for all $D_A(h) \leq w$, with compatibility under basic-open restriction. The construction is the module analogue of the existing construction of the affine structure sheaf. Evaluation on the component h gives

$$\widetilde{N}(D_A(h)) \cong N_h. \tag{6}$$

Its inverse restricts an element of N_h to every basic open below $D_A(h)$. The inverse laws are componentwise compatibility and uniqueness of localisation maps. For a general finite cover, refine inside basic opens and use Lemma 3.4 for the resulting cover of each basic open. This gives sheafhood. Ring operations, module operations, and scalar restriction are componentwise. A module map induces a sheaf morphism by its localisations. These observations also prove

$$\widetilde{N}|D_A(h) \cong \widetilde{N_h} \quad \text{over } A_h.$$

Definition 4.2. A module sheaf M is quasi-coherent if there merely exist a finite affine cover w_i of X and modules N_i over $\mathcal{O}(w_i)$ with $M|w_i \cong \widetilde{N_i}$. No chosen atlas or chosen N_i is part of the property.

Restriction of a quasi-coherent module to a compact open is quasi-coherent: refine its intersections with the given w_i by principal opens inside w_i and use the last displayed comparison. The same argument proves locality: if a finite compact-open cover has quasi-coherent restrictions, collecting their finite affine witnesses gives witnesses for Definition 4.2. Both conclusions are propositions, so the atlas choices may be eliminated. Neither fact is used in the proof of Theorem 1.1.

Canonical maps first; models only to prove equivalence. For any module sheaf M, any $a \in L$ and $g \in \mathcal{O}(a)$, set $a_g = D_a(g)$. Restriction gives an $\mathcal{O}(a)$-linear map $M(a) \to M(a_g)$, with target scalars restricted along $\mathcal{O}(a) \to \mathcal{O}(a_g)$. Since $g|a_g$ is a unit, its action on $M(a_g)$ is invertible. The module-localisation universal property therefore gives a specified map

$$\lambda_{a,g} : M(a)_g \longrightarrow M(a_g), \qquad \lambda_{a,g}(x/g^r) = (g|a_g)^{-r}(x|a_g). \tag{7}$$

No quasi-coherence or local model is used in this definition. For $b \le a$, restriction and localisation give a commuting square

$$\begin{array}{ccc} M(a)_g & \xrightarrow{\lambda_{a,g}} & M(D_a(g)) \\ \downarrow & & \downarrow \\ M(b)_{g|b} & \xrightarrow{\lambda_{b,g|b}} & M(D_b(g|b)). \end{array}$$

Indeed, after restriction of scalars to $\mathcal{O}(a)$ both composites are linear extensions of the same restriction on original sections; the action of g is invertible in the target. Uniqueness of localisation extensions proves their equality. The same argument gives compatibility when one basic open $D(h)$ is contained in another $D(g)$: the action of g is already invertible on $D(h)$. Thus every comparison diagram below is built from canonical maps which already commute. Local models are used only to prove the components are equivalences.

Lemma 4.3 (The λ-criterion)**.** *Let w be affine and put $B = \mathcal{O}(w)$. The following are equivalent: $M|w$ is associated to a B-module; and for every $s \in B$ the canonical map*

$$\lambda_{w,s} : M(w)_s \longrightarrow M(D_w(s))$$

is an isomorphism. Under the second condition the maps extend uniquely to a natural linear isomorphism $\widetilde{M(w)} \cong M|w$, along the canonical affine chart. Under the first, more generally $\lambda_{v,s|v}$ is an isomorphism for every principal open $v = D_w(h)$ and every $s \in B$.

Proof. Both directions use the element form of a localisation equivalence. For a B-linear $u : P \to Q$ such that f acts invertibly on Q, the extension $P_f \to Q$ is bijective if and only if u *clears* (every $y \in Q$ merely has $f^r y = u(x)$) and *kills* ($u(x) = 0$ implies merely $f^r x = 0$). Both conditions are elementwise, so they pass along a commuting square whose sides are bijective

B-linear maps, and they are unchanged when an $\mathcal{O}(v)$-module is regarded as a B-module along $B \to \mathcal{O}(v)$, with s in place of its image.

Let e be a model, $M|w \cong \widetilde{N}$ along the chart π, let $v = D_w(h) = \pi(D_B(h))$ and put $t = sh$. Then $D_v(s|v) = v \wedge D_w(s) = D_w(sh) = \pi(D_B(t))$. Evaluating e on these basic opens gives bijections $i : M(v) \cong N_h$ and $j : M(D_v(s|v)) \cong N_t$. They are B-linear: the chart sends the section $b/1$ to $b|$, e is linear over the structure sheaf, and evaluation is semilinear over ring evaluation. Naturality of e and of evaluation turns restriction $\rho : M(v) \to M(D_v(s|v))$ into the restriction $N_h \to N_{sh}$, that is $j\rho = ri$. The map r clears and kills at s: the restriction of x/h^p is $s^p x/(sh)^p$, and a fraction vanishing in N_{sh} is killed by a power of s in N_h. Hence ρ clears and kills, and $\lambda_{v,s|v}$ is an isomorphism. The case $h = 1$, $v = w$, is the first direction. Affineness of w is not used.

Conversely, evaluate $\widetilde{M(w)}$ on the principal basis and use $\lambda_{w,s}$ as its comparison with M. The squares proved above give naturality, including arbitrary inclusions between basic opens. Regard the maps as linear maps over the structure sheaf using the canonical basic-ring comparisons. The basis-map extension theorem constructs their unique global morphism. The local-equivalence theorem, using finite basic covers, makes that morphism an equivalence. No section-localisation theorem or reconstruction result is assumed. □

Consequently Definition 4.2 is equivalently the mere existence of a finite affine cover on whose members all canonical λ maps are isomorphisms. We retain the associated-model definition, but use this criterion to supply its models. Finite choices of models or covers remain under truncation until the quasi-coherence property is obtained.

Lemma 4.4 (Finite sums)**.** *Finite direct sums of module sheaves are formed sectionwise and are sheaves. On an affine w, if $M_i|w \cong \widetilde{N_i}$, then*

$$(\bigoplus_i M_i)|w \cong \widetilde{\bigoplus_i N_i}.$$

The empty sum is the zero sheaf.

Proof. For a finite family, sums are products. Compatible tuples glue coordinatewise, including the empty tuple. Localisation commutes with finite products by choosing common denominator bounds; the canonical maps for the sum are the coordinatewise canonical maps. Apply the λ-criterion, or equivalently the same basic comparisons, to obtain the displayed associated model. Its coefficient module is the finite sum of the N_i, using their top evaluations. □

Theorem 4.5 (Descent of canonical localisation)**.** *Let u_i be a finite cover of X, M a module sheaf and $g \in A = \mathcal{O}(1)$. Suppose the canonical localisation maps on every member u_i and every overlap $u_i \wedge u_j$, at the respective restrictions of g, are isomorphisms. Then*

$$\lambda_{1,g} : M(1)_g \longrightarrow M(D(g))$$

is an isomorphism of A_g-modules. Neither affineness nor associated models are assumptions of this pure descent statement.

Proof. The sheaf condition expresses $M(1)$ as the equaliser

$$M(1) \longrightarrow \prod_i M(u_i) \rightrightarrows \prod_{i,j} M(u_i \wedge u_j).$$

Regard every term as an A-module through restriction. Localise at g and compare with the sheaf equaliser for the cover of $D(g)$ by $u_i \wedge D(g)$. All maps are the canonical maps defined before any models; their squares commute by the localisation universal property. Localisation

over A agrees with localisation at the image of g in each section ring, after explicit restriction of scalars. The target diagram is also regarded as an A-module diagram, by restriction from $\mathcal{O}(D(g))$.

The finite-equaliser localisation descent lemma, with the assumed member and overlap equivalences, makes the middle canonical map an isomorphism. Its algebraic input is preservation of finite products and equalisers under localisation, the relevant part of Lemma 3.1. No additional kernel or associated-exactness calculation is required here. The empty-cover sheaf condition gives zero modules on the empty space and includes that case. □

Corollary 4.6 (Canonical localisation on an affine)**.** *Let X be affine, $A = \mathcal{O}(1)$ and M quasi-coherent. Then $\lambda_{1,g} : M(1)_g \to M(D(g))$ is an isomorphism for every $g \in A$. Consequently $M \cong \widetilde{M(1)}$ along the canonical chart (affine reconstruction).*

Proof. The claim is a proposition; merely take quasi-coherent witnesses w_j with models. The canonical lattice map $\mathrm{Zar}(A) \to L$ is bijective, so the preimages of the w_j cover $\mathrm{Zar}(A)$ and are merely refined by finitely many $D_A(f_k)$, with $D(f_k) \leq w_{j(k)}$ and $\bigvee_k D(f_k) = 1$. The members $D(f_k)$ and the overlaps $D(f_k) \wedge D(f_l) = D(f_k f_l)$ are principal opens $D(f)$ contained in the chart $w = w_{j(k)}$, and such an open is the principal open $D_w(f|w)$ of the chart. By the λ-criterion, in its general form, λ at $D(f)$ at the restriction of g is an isomorphism, since $g|D(f) = (g|w)|D(f)$. Theorem 4.5 gives $\lambda_{1,g}$. The top open is affine, so the λ-criterion at $w = 1$ gives the model. No restriction of models or of quasi-coherence to smaller opens is used. □

Corollary 4.7 (Canonical localisation on every open)**.** *If M is quasi-coherent, then $\lambda_{w,g}$ is an isomorphism for every open w and every $g \in \mathcal{O}(w)$. Hence on every affine open w, $M|w \cong \widetilde{M(w)}$ along the chart.*

Proof. Say M is good at w when every $\lambda_{w,g}$ is an isomorphism. (1) Let w be a chart with a model and $v = D_w(h)$. Every $c \in \mathcal{O}(v)$ is $e\,(x|v)$ with e a unit: the ring map $B_h \to \mathcal{O}(v)$ is bijective, so c is the image of some x/h^p, and $h^p c = x|v$ with $h|v$ a unit. The λ-criterion gives $\lambda_{v,x|v}$; a unit factor changes neither D_v nor clearing and killing, so $\lambda_{v,c}$ is an isomorphism. (2) If $u \leq w$ for such a chart, the preimage of u in $\mathrm{Zar}(B)$ is merely $D_B(h_1) \vee \cdots \vee D_B(h_n)$; so u is covered by the chart opens $D_w(h_i)$, whose meets $D_w(h_i h_k)$ are chart opens, and pure descent on $X|u$ shows M good at u. (3) For any w and an atlas w_j, the $w \wedge w_j$ cover w, and they and their meets lie below the charts: descent on $X|w$ and (2). The model on an affine w is the λ-criterion. Each descent is on a slice, and λ of an open of a slice is λ of that open in X. □

For the direct implication, apply pure descent to $\mathcal{O}$ using only affine chart ring facts. The corollary is used in the converse, where the model it supplies reduces affine torsor triviality to associated modules. No compact-open localisation lemma or reconstruction sheaf map is needed for either application.

Optional module theory. The following results are consequences, not prerequisites for the criterion or for affine torsor triviality.

Lemma 4.8 (Exactness for associated modules)**.** *For an A-linear map $u : N \to P$, there are canonical comparisons*

$$\widetilde{\ker u} \cong \ker(\widetilde{u}), \qquad \widetilde{\operatorname{coker} u} \cong \operatorname{coker}_{\mathrm{sheaves}}(\widetilde{u}).$$

The map $\widetilde{P} \to \widetilde{\operatorname{coker} u}$ is locally surjective. In particular, for an ideal $I \subseteq A$, the kernel of $\widetilde{A} \to \widetilde{A/I}$ is $\widetilde{I}$.

Proof. The sectionwise kernel of a module-sheaf map is a sheaf: glue in the source and check the kernel equation by separation in the target. On $D_A(h)$, Lemma 3.1 identifies $(\ker u)_h$ with $\ker(u_h)$. Evaluation (6) gives the comparison with the sheaf kernel there. These maps commute with restriction, since they commute with the kernel inclusions into N_h. Extend over finite basic covers by sheaf gluing and separation to obtain the first comparison.

For the quotient, finite exactness identifies $(\operatorname{coker} u)_h$ with $\operatorname{coker}(u_h)$. An element of this quotient merely lifts to P_h. Thus the map to the associated quotient is locally surjective, by a finite basic refinement of any compact open. A sheaf morphism from $\widetilde{P}$ which kills $\widetilde{u}$ factors uniquely on each basic open through this module quotient; uniqueness makes the factors natural, and the sheaf condition extends them to all opens. This proves the sheaf cokernel property. For the last assertion apply the kernel comparison to $A \to A/I$, whose module kernel is I. □

This lemma concerns maps already supplied as associated maps. It requires neither affine reconstruction nor localisation of sections on a general compact open.

Corollary 4.9. *On an affine $X = \operatorname{Spec} A$, every quasi-coherent module M is canonically $\widetilde{M(1)}$. A morphism between such sheaves is determined on basic opens by the localisations of its global map.*

Proof. Corollary 4.6 gives the comparison on every basic open, with its restriction equations. Extend it to all opens using their finite basic covers and the sheaf condition. For a morphism, linearity and naturality force the value on n/f^r to be the image of n divided by f^r; these fractions exhaust the localised module. □

Lemma 4.10 (Kernels and quotient sheaves)**.** *The kernel of a morphism between quasi-coherent modules is quasi-coherent. For a module map $N \to P$ on an affine, the sheaf $\widetilde{\operatorname{coker}(N \to P)}$ is its sheaf cokernel. The map to this cokernel is locally surjective.*

Proof. The cokernel assertions are Lemma 4.8. For the kernel, take finite affine witnesses for the two quasi-coherent sheaves and a common finite affine refinement. On that refinement, Corollary 4.9 identifies both sheaves and the morphism with their associated global modules and map. Lemma 4.8 then identifies the sheaf kernel with the associated module kernel on each member. These models give quasi-coherence. This general closure result uses reconstruction; the partition kernel in the direct proof does not. □

5. Torsors and first cohomology

The torsor definition below is used in the hypothesis and the lifting lemma. The results of this section establish the converse and the cohomological reformulation; the direct implication uses no affine torsor-triviality or classification theorem.

An M-torsor T is a sheaf of sets with an action of the additive sheaf M. For every a, whenever $t, t' \in T(a)$ there is a unique $m \in M(a)$ with $t' = t + m$; and every a merely has a finite cover on which T has sections. Equivalently, the action-difference map $M \times T \to T \times T$ is an isomorphism and T is locally inhabited. A global section identifies T with M by translation. An equivariant morphism of torsors is automatically an isomorphism, as is checked after a local trivialisation and then by sheafhood.

Lemma 5.1 (Torsors under an associated module on an affine)**.** *For an A-module N, every torsor under $\widetilde{N}$ on $\operatorname{Spec} A$ merely has a global section.*

Proof. Local inhabitation at the top gives a finite compact-open cover with sections. Refine it by finitely many principal opens $D_A(f_i)$ and restrict the sections to obtain $t_i \in T(D_A(f_i))$, with $\bigvee_i D_A(f_i) = 1$.

On each overlap write $t_i = t_j + c_{ij}$. Evaluation at $f_i f_j$ gives $c_{ij} \in N_{f_i f_j}$, and uniqueness of differences gives the cocycle equation on triple overlaps. Apply Lemma 3.5 directly with the lattice covering equation to obtain, merely, $b_i \in N_{f_i}$ with $b_j - b_i = c_{ij}$. Extend b_i to the corresponding associated section on $D_A(f_i)$. The corrected sections $t_i + b_i$ agree on overlaps and therefore glue. All choices remain inside the mere existence assertion. If the cover is empty, the same sheaf gluing step supplies the unique section. No separate conversion of the cover to a unit-ideal witness is needed in this torsor argument; that algebra is internal to degree-one gluing. □

Lemma 5.2 (Torsors along a chart)**.** *Let w be affine, $B = \mathcal{O}(w)$, and e a model of $M|w$: an isomorphism from the direct image of $M|w$ along the chart $\pi : \mathrm{Zar}(B) \to L$ to $\widetilde{N}$. For an M-torsor T put $T'(V) = T(\pi V)$, with $t + x = t + e^{-1}(x)$. Then T' is a $\widetilde{N}$-torsor on $\operatorname{Spec} B$, and a global section of T' is a section of T over w.*

Proof. The chart preserves finite joins and meets, so a cover of V gives one of πV, compatibility transfers along the meet equation, and the sheaf condition is that of T. The action laws are those of T, since e^{-1} is additive and natural. The difference of t, u is $e(u-t)$, uniquely, because e is bijective. For local sections, cover πV by opens with sections; the chart is bijective since w is affine, so these opens are images of opens of $\operatorname{Spec} B$, and since π preserves meets and is injective, the order of $\operatorname{Spec} B$ is reflected: the preimages lie below V and cover it. Finally $\pi(1) = w$. □

Corollary 5.3 (Affine torsor triviality)**.** *Every torsor under a quasi-coherent module on an affine scheme merely has a global section. More generally, on any X, every torsor under a quasi-coherent module merely has a section over every affine open.*

Proof. By Corollary 4.6, M has a model $M \cong \widetilde{M(1)}$ along the chart of the top open. Move the torsor along it (Lemma 5.2) and apply Lemma 5.1. The empty case is included. For an affine open w, use instead the model at w of Corollary 4.7; Lemma 5.2 then gives a section over w. □

Fix a finite cover $\mathcal{U} = (u_i)$ of X. Define $Z^1(\mathcal{U}, M)$ to be families $c_{ij} \in M(u_i \wedge u_j)$ with $c_{ij} + c_{jk} = c_{ik}$ after restriction to triple overlaps. These equations imply $c_{ii} = 0$ and $c_{ji} = -c_{ij}$. For $b_i \in M(u_i)$ put $(\delta b)_{ij} = b_j|u_i - b_i|u_j$. Define $c \sim c'$ when there merely exist b_i with $c' = c + \delta b$. This is an equivalence relation compatible with addition, so the set quotient

$$\check{\mathrm{H}}^1(\mathcal{U}, M) = Z^1(\mathcal{U}, M)/\sim$$

is an abelian group. The cocycle type and this quotient are small.

Proposition 5.4 (Cocycles and torsors)**.** *The group $\check{\mathrm{H}}^1(\mathcal{U}, M)$ classifies, up to equivariant isomorphism, the M-torsors which merely have sections on every u_i. For such torsors, zero corresponds exactly to those which merely have a global section.*

Proof. For $c \in Z^1(\mathcal{U}, M)$, define $T_c(a)$ to be families

$$x_i \in M(a \wedge u_i), \qquad x_j - x_i = c_{ij}|(a \wedge u_i \wedge u_j).$$

Restriction and the $M(a)$-action are componentwise. This is a sheaf: glue each coordinate and use separation for the displayed equations. If $a \leq u_i$, the coordinate x_i determines all others by $x_j = x_i|u_j + c_{ij}|(a \wedge u_j)$. The cocycle law proves compatibility. Thus $T_c|u_i$ is trivial. For

any two points, their coordinate differences agree on overlaps and glue to the unique element of $M(a)$ translating one to the other. Hence T_c is a torsor. The local point with i-coordinate zero will be denoted t_i; its differences satisfy $t_i = t_j + c_{ij}$.

Replacing c by $c + \delta b$ gives the equivariant isomorphism $(x_i) \mapsto (x_i + b_i)$. Conversely any equivariant isomorphism between two such torsors is translation by a unique b_i in their local coordinates; its compatibility equations give exactly the coboundary relation. Consequently the quotient identifies precisely the isomorphism classes, not any extra pairs.

For a torsor T with sections on every u_i, finite choice gives a mere family t_i. Their differences $t_i = t_j + c_{ij}$ give a cocycle. A point $t \in T(a)$ has coordinates x_i defined by $t|u_i = t_i + x_i$; then $x_j - x_i = c_{ij}$. This identifies T with T_c. Changing the local sections changes the cocycle by a coboundary, so its quotient class is uniquely determined and can be extracted from the mere family of sections. If T has a global section, its coordinates show the class is zero; if the class is zero, the coboundary equation adjusts the t_i to a compatible family and glues a global section, merely. □

Corollary 5.5 (The H^1 form)**.** *Let $\mathcal{U} = (u_i)$ be a finite cover of X and M a module sheaf.*

1. *If every M-torsor merely has a global section, then $\check{\mathrm{H}}^1(\mathcal{U}, M) = 0$.*
2. *If $\check{\mathrm{H}}^1(\mathcal{U}, M) = 0$, every M-torsor with sections on the members u_i merely has a global section.*
3. *If $\mathcal{U}$ is affine and $\check{\mathrm{H}}^1(\mathcal{U}, K) = 0$ for every quasi-coherent K, then X is affine.*

Hence, for a qcqs X, the following are equivalent: X is affine; $\check{\mathrm{H}}^1(\mathcal{U}, M) = 0$ for every finite affine cover $\mathcal{U}$ and every quasi-coherent M; this holds for one merely given finite affine cover.

Proof. (1) Every class is that of a cocycle c; the class of T_c is c, and a torsor with a global section has class zero. (2) The class of the torsor is zero; apply Proposition 5.4. (3) The partition torsor of Lemma 6.1 for the cover $\mathcal{U}$ has points on the members: the basis vector of 1 on u_j. Its translation module is quasi-coherent (Proposition 8.1), so by (2) the torsor merely has a global section, which is a partition of unity subordinate to $\mathcal{U}$; conclude as in Sections 8 and 9. For the equivalence, an affine X satisfies the torsor hypothesis by Corollary 5.3, so (1) applies to every cover; a qcqs X merely has a finite affine cover; and (3) concludes. □

Thus the theorem is stated by vanishing, and its H^1 form needs no comparison of the groups of two covers. That comparison nevertheless holds, through torsors.

Lemma 5.6 (Refinement)**.** *Let $\mathcal{W} = (w_a)$ refine $\mathcal{U}$, with $w_a \leq u_{\tau a}$. Restriction $\rho(c)_{ab} = c_{\tau a \tau b}|(w_a \wedge w_b)$ induces an additive map $\check{\mathrm{H}}^1(\mathcal{U}, M) \to \check{\mathrm{H}}^1(\mathcal{W}, M)$ which sends the $\mathcal{U}$-class of every torsor to its $\mathcal{W}$-class. If every M-torsor has sections on the members of $\mathcal{U}$, it is bijective.*

Proof. The restricted cochain is a cocycle: restrict the cocycle law of $\mathcal{U}$ to the triple overlap. For sections t_i on $\mathcal{U}$, the restricted sections $t_{\tau a}|w_a$ have cocycle ρ of that of the t_i, since differences commute with restriction. In particular $\rho(c)$ is the cocycle of the restricted canonical points of T_c. If $c \sim d$, then $T_c \cong T_d$ (Proposition 5.4 on $\mathcal{U}$), so their $\mathcal{W}$-classes agree; this makes the map well defined, and additivity is that of restriction. Under the hypothesis, send $[d]$ to the $\mathcal{U}$-class of T_d. Its image is the $\mathcal{W}$-class of T_d, that is $[d]$; and $T_{\rho(c)}$, T_c have the same $\mathcal{W}$-class, so are isomorphic and have the same $\mathcal{U}$-class $[c]$. □

Corollary 5.7 (Independence of the affine cover)**.** *For a quasi-coherent M and finite affine covers $\mathcal{U}$, $\mathcal{V}$, there is a canonical isomorphism $\check{\mathrm{H}}^1(\mathcal{U}, M) \cong \check{\mathrm{H}}^1(\mathcal{V}, M)$ of groups. It sends the $\mathcal{U}$-class of every torsor to its $\mathcal{V}$-class, in particular $[c]$ to the class of T_c, and its inverse is the comparison from $\mathcal{V}$ to $\mathcal{U}$.*

Proof. Let $\mathcal{Y}$ be the concatenation of the two families; $\mathcal{U}$ and $\mathcal{V}$ refine it with equal members. Every torsor has sections on the affine members of $\mathcal{Y}$ (Corollary 5.3), so Lemma 5.6 gives additive bijections $\check{\mathrm{H}}^1(\mathcal{Y}) \to \check{\mathrm{H}}^1(\mathcal{U})$ and $\check{\mathrm{H}}^1(\mathcal{Y}) \to \check{\mathrm{H}}^1(\mathcal{V})$, both sending the class of a torsor to its class. Compose the second with the inverse of the first. No contracted sum of torsors and no common refinement by intersections is needed, and the restriction of quasi-coherence to affine opens is replaced by Corollary 4.7. □

The atlas-free proposition

$$\mathrm{Acyclic}_1(X, M): \quad \text{every } M\text{-torsor merely has a global section}$$

is the hypothesis of the formal proof. This avoids extracting a chosen atlas from the qcqs property in order to state the hypothesis. The classification uses a small Cech quotient to represent torsor classes; it does not postulate a small type of all module sheaves or all torsors. Quantifying over those structures may live one universe higher, as in the existing geometric category construction.

No affineness is used in Proposition 5.4, and only the members of $\mathcal{U}$ in Corollary 5.5(3). The pair and triple intersections carry section groups and equations; their affineness and their higher acyclicity are not used.

6. The fibre of an addition map

Lemma 6.1 (Partition torsor)**.** *Let $u_1, \dots, u_m$ be a finite cover and let J_i be sheaves of ideals of $\mathcal{O}$ such that $J_i|u_i = \mathcal{O}|u_i$. Define sectionwise*

$$E = \bigoplus_i J_i, \qquad K(a) = \{(t_i) \in E(a) \mid \sum_i t_i = 0\}, \qquad T(a) = \{(t_i) \in E(a) \mid \sum_i t_i = 1\}.$$

Then K is a module sheaf and T is a K-torsor. If K is quasi-coherent, the hypothesis of Theorem 1.1 gives, merely, $t_i \in J_i(1)$ with $\sum_i t_i = 1$.

Proof. Finite tuples glue coordinatewise. Each displayed equation can be checked locally by separation in $\mathcal{O}$, so both fibres are sheaves. The homogeneous equation defining K is preserved by the sectionwise module operations, making K a module sheaf. Addition of a zero-sum tuple acts on T. For $t, t' \in T(a)$, the unique translating element is $t' - t \in K(a)$. On u_i, the tuple with ith coordinate 1 and all others 0 is a section of T. Restrictions of these sections to $a \wedge u_i$ prove local inhabitation on every a. Thus T is a K-torsor, and the atlas-free hypothesis applies once K is quasi-coherent. For $m = 0$ the cover implies $1_L = 0_L$; the empty-cover sheaf condition gives $1 = 0$ in the section ring and the unique empty tuple. □

Only these local sections of this particular fibre are needed. No theorem about local surjectivity on arbitrary targets is required. The existing general sheaf-of-lifts construction may be reused for this special case, but it is not a further mathematical prerequisite.

7. The support ideal

Fix $U \in L$. Define

$$J_U(a) = \{t \in \mathcal{O}(a) \mid D_a(t) \le U\}. \tag{8}$$

The section t is the underlying element; membership is a proposition.

Lemma 7.1 (Ideal sheaf)**.** *J_U is a sheaf of ideals in $\mathcal{O}$ and $J_U|U = \mathcal{O}|U$ canonically.*

Proof. Zero belongs by (2). If s, t belong, then $D_a(s+t) \le D_a(s) \vee D_a(t) \le U$; negatives belong because $D_a(-t) = D_a(t)$. For $r \in \mathcal{O}(a)$, $D_a(rt) = D_a(r) \wedge D_a(t) \le U$. Thus $J_U(a)$ is an ideal. Equation (1) proves restriction stability. For a compatible family of ideal sections, first glue in $\mathcal{O}$; equation (4) proves that the glued section belongs to J_U. Uniqueness follows from uniqueness in $\mathcal{O}$, because ideal membership is a proposition. If $a \le U$, every $D_a(t)$ is below $a \le U$, so the inclusion $J_U(a) \to \mathcal{O}(a)$ is bijective, compatibly with all operations and restrictions. □

Lemma 7.2 (Saturation)**.** *Let $S \subseteq E$ be a submodule sheaf, $a \in L$ and $g \in \mathcal{O}(a)$, and suppose $\lambda^E_{a,g}$ is an isomorphism. Suppose S is* saturated at g*: every $x \in E(a)$ with $x|D_a(g) \in S(D_a(g))$ merely has some e with $g^e x \in S(a)$. Then $\lambda^S_{a,g}$ is an isomorphism.*

Proof. Write ι for the inclusion. Its localisation ι_g is injective: if $\iota(x)/g^r = 0$ then $g^n x = 0$ in $E(a)$, hence in $S(a)$. By definition of the canonical maps, $\lambda^E \circ \iota_g = \iota \circ \lambda^S$, so λ^S is injective. For $y \in S(D_a(g))$ write $\iota y = \lambda^E(x/g^r)$. Then $x|D_a(g) = g^r\, \iota y$ lies in S, so merely $g^e x \in S(a)$, and $\lambda^S((g^e x)/g^{r+e})$ has image $\lambda^E(x/g^r) = \iota y$. Injectivity of ι gives $\lambda^S((g^e x)/g^{r+e}) = y$. The fibre is a proposition, so the mere exponent suffices. □

Lemma 7.3 (Localisation of the support ideal)**.** *For every a and $g \in \mathcal{O}(a)$, if $\lambda^{\mathcal{O}}_{a,g}$ is an isomorphism then so is $\lambda^{J_U}_{a,g}$. In particular J_U satisfies the λ-criterion on every affine, and is quasi-coherent.*

Proof. $J_U \subseteq \mathcal{O}$ is saturated with exponent 1: if $x \in \mathcal{O}(a)$ and $D_{D_a(g)}(x|D_a(g)) \le U$, then

$$D_a(gx) = D_a(g) \wedge D_a(x) = D_{D_a(g)}(x|D_a(g)) \le U$$

by (3) and (1), so $gx \in J_U(a)$. Apply Lemma 7.2. On an affine w, $\lambda^{\mathcal{O}}_{w,s}$ is an isomorphism by the chart ring fact, and Lemma 4.3 gives the model of $J_U|w$ with coefficients $J_U(w)$. □

No presentation of $w \wedge U$, radical ideal, formal Nullstellensatz or radical localisation is needed. (One can show $J_U(w) = \sqrt{I_w}$ for a presentation $w \wedge U = \bigvee D_w(h_i)$, but the proof does not use it.)

8. A subordinate partition of unity

Proposition 8.1. *Under the hypothesis of Theorem 1.1, for every actual finite affine cover $u_1, \dots, u_m$ there merely exist $f_i \in A = \mathcal{O}(1)$ with*

$$\sum_i f_i = 1, \qquad D(f_i) \le u_i. \tag{9}$$

Consequently the $D(f_i)$ form a finite global principal affine cover, and the $D_A(f_i)$ cover $\operatorname{Spec} A$*.*

Proof. Take $J_i = J_{u_i}$. Lemma 7.1 gives $J_i|u_i = \mathcal{O}|u_i$, so form the partition torsor T and its translation sheaf K from Lemma 6.1.

The translation sheaf is saturated. Use the given cover itself as the quasi-coherence atlas. Let $E = \prod_i J_i$ and $q : E \to \mathcal{O}$ the sum, so $K = \ker q$. Fix an affine member $w = u_j$ and $s \in \mathcal{O}(w)$. By Lemma 7.3 and the chart ring fact every $\lambda^{J_i}_{w,s}$ is an isomorphism, so $\lambda^E_{w,s}$ is one, coordinatewise. The kernel is saturated: if $x \in E(w)$ and $q(x)|D_w(s) = q(x|D_w(s)) = 0$, then $\lambda^{\mathcal{O}}(q(x)/1) = 0$, so injectivity of $\lambda^{\mathcal{O}}_{w,s}$ gives $s^e q(x) = q(s^e x) = 0$ for some e, that is $s^e x \in K(w)$. Lemma 7.2 makes $\lambda^K_{w,s}$ an isomorphism, and Lemma 4.3 gives a model of $K|u_j$ with coefficients $K(u_j)$. These models prove quasi-coherence of K on the original atlas. There is no splitting, no module-kernel calculation and no atlas refinement in this step.

One application of the torsor hypothesis. A global point of T, supplied merely by the hypothesis, is exactly a tuple $f_i \in J_i(1)$ with $\sum_i f_i = 1$. Its memberships give $D(f_i) \leq u_i$. The support laws imply

$$1 = D(\sum_i f_i) \leq \bigvee_i D(f_i) \leq 1.$$

Moreover

$$D(f_i) = u_i \wedge D(f_i) = D_{u_i}(f_i|u_i)$$

is a principal open of an affine, hence affine. The same unit equation in A gives $\bigvee_i D_A(f_i) = 1$ in its Zariski lattice. For the empty cover the unique empty tuple gives $1 = 0$ in A; X and $\operatorname{Spec} A$ are both empty. All conclusions remain under the mere existence assertion. □

This simultaneous lift replaces the quotient-supported construction, affine replacement, its finite induction and the later row-map lift. The only quasi-coherence argument for its translation sheaf is saturation (Lemma 7.2), used twice.

9. From that cover to affineness

We first isolate the structure-ring comparison needed for the final canonical map. Its hypotheses are the particular cover just constructed.

Lemma 9.1 (Ring localisation on a principal affine cover)**.** *Suppose $\bigvee_i D(f_i) = 1$, with each $D(f_i)$ affine and $f_i \in A = \mathcal{O}(1)$. Then for each member of the cover the canonical ring map*

$$\lambda_k : A_{f_k} \longrightarrow \mathcal{O}(D(f_k))$$

is an isomorphism.

Proof. Fix k and use Theorem 4.5 for $M = \mathcal{O}$ and $g = f_k$. Each member $U_i = D(f_i)$ is affine. Each overlap

$$U_i \wedge U_j = D_{U_i}(f_j|U_i)$$

is a principal open of an affine, hence affine too. On either of these affines w, the canonical chart ring fact

$$\mathcal{O}(w)_{f_k|w} \cong \mathcal{O}(D_w(f_k|w))$$

is exactly the required component λ. No associated model of $\mathcal{O}$ is used. Pure descent makes the global canonical module map an isomorphism.

Restriction $A \to \mathcal{O}(D(f_k))$ is a ring map inverting f_k, so it has a unique ring extension from A_{f_k}. Its underlying module map is the same canonical localisation map, by uniqueness on original elements. Hence this specified ring extension is a ring isomorphism. □

Merely choose a finite affine atlas and apply Proposition 8.1. Its single partition-of-unity equation already gives both covers: $D(f_i)$ of X and $D_A(f_i)$ of $\operatorname{Spec} A$, where $A = \mathcal{O}(1)$. Apply the preceding lemma to obtain

$$A_{f_i} \cong \mathcal{O}(D(f_i)) \tag{10}$$

by the canonical ring extensions of restriction. The canonical geometric map $\gamma : X \to \operatorname{Spec} A$ has inverse-image map $D_A(f) \mapsto D(f)$, and its ring map on these opens is precisely (10). Since $D(f_i)$ is affine, this identifies the restricted map

$$\gamma_i : X|D(f_i) \longrightarrow (\operatorname{Spec} A)|D_A(f_i)$$

with an isomorphism of affine schemes. Agreement of the canonical map with the spectrum of the ring map follows from the affine universal property; it is not an unmentioned computation.

Lemma 9.2 (Isomorphisms are local on the target)**.** *Let $g : Y \to Z$ be a morphism of locally ringed lattices and z_i a finite cover of Z. If each map $Y|g^*z_i \to Z|z_i$ is an isomorphism, then g is an isomorphism.*

Proof. For $y \in L_Y$, let $b_i \leq z_i$ be the inverse image of $y \wedge g^*z_i$ under the lattice isomorphism on the ith restriction. On intersections these inverse lattice maps agree, because both invert the restricted g^*; hence $b_i \wedge z_j = b_j \wedge z_i$. Put $b = \bigvee_i b_i$. Distributivity gives $b \wedge z_i = b_i$, and therefore $g^*b = y$, since the g^*z_i cover Y. Similarly equality of g^*b and g^*b' implies equality after meeting with every z_i, hence $b = b'$. Thus g^* is a lattice isomorphism.

For every open z, its cover by $z \wedge z_i$ and the source cover by their inverse images give sheaf equalisers. The maps on all member and overlap rings are isomorphisms by the local isomorphism hypotheses and restriction. These induce an inverse on the equaliser ring map. The inverses are natural by uniqueness, and form the inverse ringed morphism. Isomorphisms preserve invertibility opens by their universal characterisation, so the inverse is local. This gives the required geometric isomorphism. □

The lemma applied to γ and the target cover given by (9) proves Theorem 1.1. All remaining witnesses are eliminated into the proposition that this specified canonical map is an isomorphism. If the cover is empty, X is the empty lattice space, A is the zero ring, and the same empty-cover argument gives its affine identification.

Conversely an affine scheme satisfies the hypothesis by Lemma 5.3. Thus this proves an equivalence between affineness and quasi-coherent first-cohomology vanishing for the schemes considered here.

10. Formalisation

Everything above, except the optional module theory of Section 4 and the restriction and locality of quasi-coherence (replaced by Corollary 4.7), is checked in `unit/sag/qcqs` (30 October 2026). No new axioms are introduced; the development rests on the Theory parameters univalence and replacement. The main statements are:

- direct implication: `SerreAffineness.Serre[L,X,qcqs,hypothesis].affine`, with hypothesis `SerrePartitionCover.On[L,X].Hypothesis` (every torsor under every small quasi-coherent module merely has a global section) and conclusion `AffineCriterion.IsAffine`;
- converse and equivalence: `SerreAffineConverse.On[L,X,affine].hypothesis`, `SerreAffineConverse.Criterion[L,X,qcqs]`;
- H^1 form: `SerreCechVanishing.On[L,X].Theorem[qcqs]` (`affine-to-every`, `every-to-some`, `some-to-affine`);
- independence of the affine cover: `SerreCechComparison.On[L,X,M,qc].Covers[U,V]` (`equivalence`, `comparison-add`, `comparison-torsor`).

The formal proof follows the text, one module per step:

canonical maps $\lambda_{a,g}$, element form	`SerreLambda`, `SerreLambdaClearing`, `SerreLambdaElementarySquare`
chart ring fact	`SerreLambdaChart`
Lemma 4.3, $\lambda \Rightarrow$ model	`SerreLambdaCriterion`
Lemma 4.3, model $\Rightarrow \lambda$	`SerreLambdaModel`
finite products	`SerreLambdaProducts`
Theorem 4.5	`SerreLocalisationDescent`, `SerreLambdaTop`, `SerreLambdaNested`
Corollary 4.6	`SerreAffineLocalisation`
Corollary 4.7	`SerreLambdaQuasiCoherent`, `SerreLambdaUnit`
Lemma 3.5	`SerreModuleCechGluing`
Lemma 5.1	`SerreAffineTorsor`
Lemma 5.2	`SerreTorsorChart`
Corollary 5.3	`SerreAffineConverse`
Proposition 5.4	`SerreCechIsoClass`
Corollary 5.5	`SerreCechVanishing`
Lemma 5.6	`SerreCechRefinement`
Corollary 5.7	`SerreCechComparison`
Lemma 6.1	`SerrePartition` (via `SerreTorsorLifts`)
Lemma 7.1	`SerreSupportIdeal`
Lemmas 7.2, 7.3	`SerreSupportIdealSaturation`, `SerreSupportIdealLambda`
Proposition 8.1	`SerrePartitionQuasiCoherent`, `SerrePartitionCover`
Lemma 9.1	`SerreRingLocalisation`
Lemma 9.2	`SerreLocalIso`
the restricted canonical map	`SerreCanonicalMap`
Theorem 1.1, direct	`SerreAffineness`

Simplifications made for the formal proof.

(1) Saturation (Lemma 7.2) gives the localisation of the support ideal and of the partition kernel; no radical description of J_U on an affine and no splitting of the kernel are used.
(2) In Lemma 9.2 the relation $b_i \wedge z_j = b_j \wedge z_i$ is not needed: surjectivity uses only $g^*(\bigvee_i b_i) = y$.
(3) The direction model $\Rightarrow \lambda$ of Lemma 4.3 is proved in the element form, on every principal open of the chart, so no restricted model and no associated restriction comparison are needed.
(4) Corollary 4.7 replaces the restriction of quasi-coherence to affine opens.
(5) Affine torsor triviality goes through affine reconstruction and Lemma 5.2 to the associated case, not through a second degree-one argument.
(6) The H^1 form of the theorem is stated by vanishing (Corollary 5.5); the comparison of the groups (Corollary 5.7) goes through torsors and the concatenation of the two covers, with additivity from restriction instead of a contracted sum of torsors.

Validation. The full server check has 390 modules and no failures (16.4 s). The independent checker, run on the whole development with opening instrumentation, checks 16,047 declarations (35.1 s). Every one of the 6,797 declarations of the 141 Serre modules has exactly one audit entry stating which definitions it may open, and every one of the 10,549 entries of the combined audit reports that nothing else was opened.

References

[1] The Stacks Project, Lemma 30.3.1, Serre's criterion for affineness, Tag 01XF. Its closed-point proof is a comparison reference, not used here.
[2] The Stacks Project, Lemma 21.10.4, Cech cohomology and torsors, Tag 0A6G. The explicit finite-cover construction in this note gives the needed degree-one comparison without derived cohomology.